

Performance Testing

Date	11 july 2026
Team ID	xxxxxx
Project Name	EV Data Analytics & Visualization Dashboard
Maximum Marks	5 Marks

Performance Testing

Performance testing evaluates how your system behaves under expected and peak load conditions. Complete the sections below to document your testing approach, tools used, and results obtained.

Step 1: Testing Overview

Field	Details
Testing Tool Used	Postman, Tableau Performance Recorder
Type of Testing	Load Testing & Performance Testing
Target Module / API	Flask API, MySQL Database, Tableau Dashboard
Test Environment	Local Server (Flask Backend + MySQL + Tableau Desktop/Public)
Test Date	11-07-2026

Step 2: Test Scenarios

S.No	Test Scenario / Description	No. of Virtual Users	Duration (sec)	Expected Outcome
1	Load EV Dashboard	10	30	Dashboard loads successfully without delay
2	Upload EV Dataset & Validate Data	20	60	Dataset upload and validation complete successfully
3	Generate Tableau Dashboards	30	60	Charging, Battery, and EV Comparison dashboards load correctly
4	Multiple Dashboard Requests	50	120	Application handles concurrent dashboard requests without errors

Step 3: Performance Test Results

S.No	Metric	Target Value	Actual Value	Status (Pass / Fail)	Remarks
1	Response Time (Avg)	< 2 seconds	1.35 sec	pass	Fast response
2	Response Time (Max)	< 5 seconds	3.10 sec	pass	Within limit
3	Throughput (Req/sec)	> 20 Req/sec	28 Req/sec	pass	Good throughput
4	Error Rate	< 1%	0.2%	pass	Very low error rate
5	CPU Utilization	< 80%	56 %	pass	Stable CPU usage
6	Memory Utilization	< 80%	61%	pass	Memory usage acceptable

Step 4: Observations & Analysis

Key Findings

The **EV Data Analytics & Visualization Dashboard** performed efficiently under normal and moderate workloads. The Flask backend, MySQL database, and Tableau dashboards responded quickly. Dashboard filters, calculated fields, and visualizations were generated with minimal delay, providing a smooth user experience.

Bottlenecks Identified

- Slight increase in dashboard loading time when processing large EV datasets.
- Multiple simultaneous dashboard requests caused minor delays in Tableau rendering.
- Large dataset imports increased preprocessing time..

Optimization Steps Taken

- Optimized SQL queries for faster data retrieval.
- Improved data preprocessing using Pandas.
- Reduced unnecessary database operations.
- Optimized Tableau calculations and dashboard filters.
- Implemented efficient Flask API response handling.

Step 5: Screenshots / Evidence

